

MSc Thesis Problem Statement: Experts Search Engine

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1 Context

In our research project, we aimed to solve the problem of finding a supervisor, for this, we:

1. Extracted data on academics and researchers at ITU from Pure, The Danish Research Portal and LearnIT
2. Mapped the data to the corresponding expert ID containing information on publications, courses and projects for each expert
3. Implemented a BM-25F retrieval and ranking algorithm which would run on free-text user queries
4. Developed a containerised REST API with FastAPI
5. Developed a containerised React SPA which could be used to test queries and display results

During the development of our research project, we came across the following challenges:

1. The LearnIT scraper did not retrieve all courses/academics data.
2. In the data expansion phase, we aimed to enrich ITU researcher profiles by linking Person records extracted from Pure to their corresponding publications in the DRP. However, instead of linking publications based on ORCID identifiers, they were linked to authors solely based on name matching, which introduced systematic errors for individuals with common Danish names, as multiple distinct authors in the DRP may share identical or highly similar names.
3. The scoring system we have developed relies primarily on lexical matching and therefore cannot capture deeper semantic relations between terms. Additionally, queries using synonyms or semantically related terminology might produce suboptimal results, and lastly, manually defined field boosts and heuristic scoring bonuses may not generalise optimally across all domains.
4. The project was not deployed for production.
5. Database design — our database was not completely normalised and in several cases the table keys were composed of several fields (mostly due to the above-mentioned name-matching of academics).
6. Testing of output — we did not implement formal testing of queries and result analysis to measure success of weights in our implementation of BM-25F.

2 State of the Art

As previously mentioned, for the development of this project, we will use existing data from Pure, The Danish Research Portal and LearnIT, in addition to other sources such as OpenAlex. These portals used in conjunction should approximate what we expect to develop, and out of all of them, Pure is probably the most complete since it is data-rich and specific to ITU academics. Moreover, it includes data on projects and publications, as well as expert interests and preferences (if these are filled in the relevant fields). However, Pure lacks algorithmic transparency, and it may have too many features since it serves too many different purposes, therefore, the user experience can be greatly improved by simply offering an alternative that expands on its dataset and is built with a specific aim in mind: finding the expert that would be the all-round best possible fit for a given search.

3 Method and Deliverables

During the course of our thesis, we plan to fix and improve on the challenges found and described in the Context section, in the following ways:

1. Fixing the LearnIT scraper so it retrieves all the available data.
2. Implementing ORCID-based matching.
3. Investigating and implementing one (or multiple) of the following search engine improvements:
 - (a) Semantic and Hybrid Retrieval; integration of semantic embeddings (using FAISS) and/or hybrid lexical-semantic retrieval.
 - (b) Configurable Search Algorithms; development of several search algorithms and the possibility to switch among them.
 - (c) Publication Weighting; introduction of additional weights on publications based on the year of publication.
 - (d) Keyword-Based Search; implementation of keyword search.
 - (e) Academic Profile Management; allowing academics to log into the portal through email verification, so that they could provide a summary about themselves, a picture, their main topics of interest, availability to calendar, and location on campus (office).
 - (f) AI-Generated Expert Matching Summaries; development of an AI model which takes a person UUID together with retrieved papers, projects, and courses as input, and generates a concise 2–3 line explanation of why a particular expert is a strong match for the search.
4. Deploying the project to production, which could include:
 - (a) Adding an on-page search field to refine and display only content related to a new query within the selected expert’s context.
 - (b) Improving pagination by implementing dynamic pagination (e.g., a sliding-window approach).
 - (c) Enriching expert profiles with additional attributes, such as availability, topics of interest, an AI-generated summary, a self-authored summary, and office location.
 - (d) Hiding project and course sections when no corresponding results exist (to reduce visual noise).

- (e) Returning only results that are relevant to the query and highlighting matched query terms in the displayed results.
 - (f) Deploying the tool for open access.
5. Improving database design and finding a solution that will drive us to have a smaller but more accurate dataset.
 6. Formally testing the output for given search queries to determine output quality.